

Trust and Safety Audit: Mid-Point Report Document

Speaking in Code: Automated Detection of Dogwhistle Language and Covert Hate Speech on Reddit

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1 Introduction

1.1 Problem Statement

Automated content moderation systems have made significant progress in detecting overt hate speech, such as direct slurs, explicit threats, and clearly derogatory language. However, malicious actors continuously adapt their linguistic tactics to evade these automated defenses. One of the most persistent and challenging evasion strategies is the use of covert hate speech. This includes coded language ("dogwhistles"), leetspeak substitutions, emoji-based encoding, Unicode homoglyphs, and specialized in-group terminology. These tactics carry deeply hateful meanings recognizable to a target audience but remain structurally opaque to standard prediction-based classifiers.

Our own preliminary analysis underscores the scale of this vulnerability: in an 8,720-item corpus collected from ten Reddit communities, Google's Perspective API flags only 2.1% of content as toxic while OpenAI's Moderation API flags 9.6% - a 4.6× gap that widens to 3.1% versus 13.3% in polarized communities. A total of 702 items are flagged by OpenAI but missed entirely by Perspective, suggesting a large class of content that existing tools handle inconsistently.

1.2 Motivation and Significance

The inability of current classifiers to detect covert hate speech creates a serious trust and safety vulnerability because it allows harmful content to remain visible while preserving the speaker's plausible deniability. This problem disproportionately affects marginalized communities, including LGBTQ+ users, racial and ethnic minorities, and religious minorities, who are often targeted through coded rhetoric that appears benign outside the relevant in-group context [7]. In practical terms, detection failures shift the burden of recognition and response onto targets themselves, forcing them to interpret, report, and endure abuse that automated systems incorrectly

classify as non-harmful. Addressing this gap is urgent from a trust and safety perspective, as the unchecked proliferation of covert hate normalizes extremist rhetoric within mainstream digital spaces.

1.3 Scope and Project Boundaries

This project conducts a targeted platform measurement audit of two widely used off-the-shelf moderation systems: Google's Perspective API and OpenAI's Moderation API. Our study focuses on English-language Reddit posts and top-level comments drawn from ten subreddits: six polarized communities (r/conspiracy, r/Firearms, r/KotakuInAction, r/PoliticalCompassMemes, r/PublicFreakout, and r/Conservative) and four control communities (r/explainlikeimfive, r/science, r/AskHistorians, and r/todayilearned). This design allows us to compare spaces where covert identity-based hostility is more likely to appear against well-moderated, topic-specific communities where hate prevalence is expected to be relatively low.

Our scope is deliberate and specific. We are auditing classifier behavior and measuring the effect of a pre-processing normalization layer. We are not training or fine-tuning a new hate-speech detector, not conducting a longitudinal study of how dogwhistles evolve over time, nor are we studying platform policy or moderation decision-making.

1.4 Project Outcomes

By the end of the semester, we aim to deliver three concrete outputs: (1) a human-annotated gold standard for overt hate, covert hate, borderline/ambiguous content, and non-hateful content, together with inter-annotator agreement metrics, (2) a comparative precision/recall/F1 evaluation of Perspective and OpenAI Moderation on overt versus covert hate, both before and after pre-processing, and (3) a lightweight normalization pipeline, accompanied by deployment-oriented recommendations for how platforms can use pre-processing

and human escalation to better handle coded hate without re-training proprietary models.

We aim to empirically demonstrate whether this targeted, non-retraining intervention can significantly reduce the detection gap. Specifically, we intend to recommend a two-stage pre-processing architecture for production moderation workflows (Unicode normalization + leetspeak decoding, followed by dogwhistle flag escalation for human review), a human-in-the-loop escalation protocol for dogwhistle-flagged content given the high false-positive rate of lexicon matching, and a proposal that API providers add coded-language subscales (e.g., an “identity_attack_coded” dimension in Perspective API) for scoring coded rather than explicit derogatory language.

2 Related Work

The foundational mechanics of our audited systems are detailed by Gorwa et al. (2020), who distinguish between matching-based approaches (such as hash databases) and prediction-based classifiers (such as toxicity scoring models) [4]. Although these automated techniques allow for moderation at scale, Gorwa et al. warn that they increase platform opacity and often fail to capture the political nuance of speech [4]. Further illustrating these technical limitations, Gröndahl et al. (2018) demonstrated that commercial tools like Google’s Perspective API are notoriously brittle; simple mechanical evasion tactics, such as inserting typos, altering word boundaries, or utilizing leetspeak, can completely break word-based classifiers and severely degrade character-based models [5].

Beyond mechanical evasion, semantic evasion via coded language presents a profound challenge. Mendelsohn et al. define dogwhistles as expressions that convey a benign meaning to a broad audience while simultaneously signaling hateful or provocative intent to a narrow in-group [6]. In their curation of a glossary containing over 300 coded terms, they empirically proved that harmful content utilizing these expressions successfully avoids detection by standard toxicity classifiers. Furthermore, Sasse et al. highlighted that this lexicon is not static; dogwhistles evolve dynamically across online habitats as communities continuously adapt to top-down moderation pressures [8].

Recent audits of commercial moderation infrastructure confirm severe vulnerabilities regarding implicit hate. For example, a 2025 audit of automated moderation tools found that filters missed up to 94% of hateful messages in specific datasets, relying overwhelmingly on explicit slur-based moderation while almost entirely missing implicit and coded hate [9].

The abuse vectors framework from Stephenson et al. also informed our conceptual framing [10]. Although their framework addresses IoT-enabled interpersonal abuse, the core insight – that harmful actors repurpose tools designed for legitimate use – applies directly to coded hate speech, where ordinary words are weaponized to carry hateful secondary meanings that automated systems cannot distinguish from benign usage. Thomas et al. provide the harm taxonomy we draw on, identifying toxic content as one of the most accessible and scalable forms of online abuse, requiring no capabilities beyond a communication channel [11]. Their finding that LGBTQ+ individuals face 1.86 times higher odds of experiencing harassment underscores the urgency of closing the detection gap for the communities most targeted by coded rhetoric. Methodologically, our project follows the red-teaming cycle discussed in the course’s Red Teaming lectures: we first probe deployed classifiers with adversarial, evasion-oriented content to expose failure modes, then propose an engineering response in the form of a normalization layer designed to address the vulnerabilities we uncover.

Although previous work successfully highlights the vulnerabilities of toxicity classifiers [5] and maps the linguistic scope of dogwhistles (Mendelsohn et al., 2023; Sasse et al., 2023) [6, 8], there is limited research on lightweight, scalable interventions that platforms can implement without retraining massive proprietary models. Our project bridges this gap. By synthesizing a dynamic lexicon with text normalization heuristics, we transition from merely auditing the detection gap to engineering an easily deployable pre-processing layer designed to catch both mechanical (leetspeak) and semantic (dogwhistles) evasion tactics before they reach the classifier.

3 Research Plan & Current Status

Guiding Research Questions: Our three research questions (stated above) guide the study: (1) quantifying the detection gap for covert hate speech in leading off-the-shelf classifiers, (2) cataloging and categorizing the evasion strategies that produce this gap, and (3) evaluating whether a custom pre-processing layer can meaningfully close it.

Harm Type: Harassment and hate speech, specifically covert, evasion-oriented hate speech that uses coded language, character substitution, or specialized group-specific terminology to target marginalized groups while evading automated detection. Peer-reviewed studies indicate that harassment and hate speech normalizes prejudice, fosters hostility, and acts as a precursor to physical discrimination and violent, extremist behavior [2, 3, 7].

Platform: Reddit, focusing on English-language communities known for politically polarized or contentious

discourse. We sampled from subreddits across a spectrum of moderation intensity and political orientation, selecting specific communities during the data collection phase based on preliminary exploration. We also included well-moderated control subreddits (e.g., r/AskHistorians, r/science) where covert hate speech is expected to be rare, to calibrate our detection tools against a stable low-prevalence baseline.

3.1 Methodological Approach

Data Collection: Our original plan called for data collection via PRAW (Python Reddit API Wrapper). During the data collection phase, Reddit API approval delays led us to pivot to the Arctic Shift public archive¹, which provides bulk access to historical Reddit post and comment data. This pivot did not affect our study design, as Arctic Shift provides the same fields we planned to collect. All data is from publicly posted content; no private messages or user account data were collected. We collected 8,720 posts and comments from ten subreddits. Six polarized subreddits were chosen for documented proximity to contentious identity-related discourse: r/conspiracy (1,197 items), r/Firearms (1,151), r/KotakuInAction (1,028), r/PoliticalCompassMemes (934), r/PublicFreakout (817), and r/Conservative (461). Four well-moderated control subreddits provide a low-prevalence baseline: r/explainlikeimfive (928), r/science (806), r/AskHistorians (740), and r/todayilearned (658). Polarized subreddits were selected for their history of coded political and identity-related speech without being dedicated hate subreddits; control subreddits are well-moderated and topic-specific with low expected hate speech prevalence.

Automated Detection Approach: Our methodology centers on a three-way comparison designed to quantify the detection gap and evaluate our proposed intervention:

- **Component 1 – Off-the-Shelf Baselines (Red-Teaming Phase):** We passed every collected post and comment through two leading, freely available content moderation APIs: Google’s Perspective API², which returns toxicity, severe toxicity, identity attack, insult, and threat scores on a 0–1 scale; and OpenAI’s Moderation API³, which classifies content across categories including hate, hate/threatening, harassment, and self-harm. Both APIs are free, require only an API key to access, and support high request volumes (Perspective API permits about 50 queries per second). For each item, we recorded all returned scores and flagged items above standard toxicity thresholds (e.g., 0.7 for Perspective API). By comparing these automated classifications against our human-annotated gold standard, we will measure the

detection gap, namely identifying the specific items that contain hate speech but that existing tools fail to flag.

- **Component 2 – Custom Normalization + Lexicon Layer (Mitigation Phase):** We built a pre-processing pipeline that normalizes evasion tactics before passing text to the same off-the-shelf classifiers. This layer has three sub-components: (a) a text normalizer that decodes leetspeak substitutions (e.g., “1” instead of “l”, “3” instead of “e”, “0” instead of “o”), strips inserted punctuation and whitespace (“h.a.t.e” instead of “hate”), and resolves Unicode homoglyphs to their ASCII equivalents, (b) a curated dogwhistle and coded-term lexicon sourced from the glossary of Mendelsohn et al. [6] (300+ terms with contextual annotations), the Anti-Defamation League (ADL) Hate Symbols Database⁴, Hatebase⁵, and supplemented by terms from Sasse et al. [8], mapping coded terms to their underlying hateful meanings, and (c) pattern-matching rules for common structural evasion patterns such as deliberate misspellings, asterisk censoring, and emoji substitutions for slurs. The normalized text was passed through the same Perspective and OpenAI Moderation APIs, and we will compare detection rates before and after normalization.

Normalization Pipeline Status: The normalization pipeline is fully implemented and has been applied to the entire corpus. It consists of three modules: (a) `homoglyphs.py`, which resolves approximately 150 Unicode codepoints (Cyrillic, Greek, fullwidth Latin) to ASCII equivalents; (b) `leetspeak.py`, a three-pass normalizer that strips interstitial punctuation, performs leet substitution with context guards, and converts asterisk-censored words to a <CENSORED> placeholder; and (c) `lexicon.py`, which pattern-matches against a curated lexicon of approximately 50 seed terms from Mendelsohn et al. and the ADL Hate Symbols Database. Of 8,720 items, 72 (0.8%) had text changed by normalization and 166 (1.9%) received dogwhistle flags, though only 31 (0.4%) had flags beyond highly ambiguous terms like “based.” The full pipeline source code is publicly available.⁶ Development was assisted by Claude Code (Anthropic), using Opus 4.6 for architectural planning and Sonnet 4.6 for implementation.

- **Component 3 – Human-Annotated Gold Standard:** Two team members independently annotate a stratified subset of 75 posts and comments. Each item will be labeled along two dimensions: (1) hate speech

¹<https://arctic-shift.photon-reddit.com/>

²<https://perspectiveapi.com>

³<https://platform.openai.com/docs/guides/moderation>

⁴<https://www.adl.org/hate-symbols>

⁵<https://hatebase.org>

⁶<https://github.com/taahamohsin/covert-speech-detection>

classification ("Overt Hate", "Covert Hate", "Borderline/Ambiguous", "Not Hateful") and (2) evasion strategy employed, if any (leetspeak, dogwhistle/coded term, deliberate misspelling, emoji substitution, Unicode homoglyph, punctuation insertion, none). This gold standard will serve as the ground truth for evaluating all automated approaches.

To measure inter-rater agreement beyond chance, we use Cohen's kappa (κ), a standard reliability metric that corrects for the agreement expected by random labeling. We adopt $\kappa \geq 0.6$ as our threshold for acceptable agreement, following conventions in computational linguistics annotation [1].

Annotation Protocol and Calibration: We designed the annotation process with methodological safeguards: Perspective API and OpenAI Moderation scores were hidden from both annotators to prevent anchoring, and annotators did not have access to each other's labels until both had completed their independent annotations.

Round 1 (75 items): The first annotation sample was stratified to capture both moderation-system agreement and disagreement, as well as baseline content from different community types. The strata were: 13 items flagged by both Perspective and OpenAI Moderation, 7 flagged only by Perspective, 20 flagged only by OpenAI Moderation, 25 flagged by neither API from polarized subreddits, and 10 flagged by neither API from control subreddits. This design allowed us to examine where the two APIs agreed, where they diverged, and how much potentially harmful content might be missed in polarized versus control contexts. Two annotators independently labeled each item for hate speech classification and evasion strategy. Cohen's kappa yielded $\kappa = 0.00$.

Annotator A classified all 75 items as "Not Hateful," while Annotator B identified 13 as Overt Hate, 4 as Covert Hate, and 7 as Borderline/Ambiguous. Although raw agreement was 68%, this was entirely attributable to base rates.

We identified two contributing factors. First, the stratification was designed to capture API agreement and disagreement patterns rather than to enrich directly for covert hate prevalence. As a result, a substantial portion of the sample consisted of items flagged by neither system, many of which were genuinely benign and may have anchored Annotator A toward a "Not Hateful" default. Second, the codebook lacked sufficient calibration examples for borderline and context-dependent cases, leading to divergent interpretation thresholds between annotators.

Round 2 (revised 50 items): Following consultation with the course assistant, we redesigned the pilot to enrich the selected items more directly for potentially covert hate-bearing content rather than API disagreement structure alone: antisemitic/conspiracy coded (6), racist/xenophobic coded (11), Islamophobic coded (4), misogynistic/incel coded (8), anti-LGBTQ/transphobic coded (7), mixed coded hate/evasion (4), and not hateful controls (10). Annotator B has completed this round, labeling 19 items as Overt Hate, 3 as Covert Hate, 4 as Borderline/Ambiguous, and 24 as Not Hateful. Annotator A's labels are pending. Once both are complete, we will compute Cohen's kappa with a target of $\kappa \geq 0.6$, adjudicate disagreements with the third teammate (who did not originally annotate the items) as tiebreaker, and expand to the full 100-item gold standard if agreement is acceptable.

Evaluation: We will compute precision, recall, and F1-score for each detection approach (Perspective API alone, OpenAI Moderation API alone, each API with our normalization layer applied) against the human-annotated gold standard. We will break down results by hate speech type (overt vs. covert) and by evasion strategy, enabling us to identify which specific strategies are most effective at defeating each classifier and which are most improved by our normalization layer.

3.2 Progress to Date

Data collection (8,720 items), Perspective API scoring (8,720/8,720), and OpenAI Moderation API scoring on raw text (8,720/8,720) are complete. The normalization pipeline is built and applied to the full corpus. OpenAI Moderation scoring on normalized text is approximately 84.9% complete (7,404/8,720), limited by API rate quotas. The dogwhistle lexicon (50 terms) and annotation codebook are finalized. Pilot annotation (75 items) is complete, revealing the calibration failure described above; a revised annotation sample is ready for re-annotation.

Preliminary findings. Perspective API flags 2.1% of all items versus OpenAI's 9.6%. In polarized subreddits, the gap widens: 3.1% versus 13.3%. A total of 702 items are flagged by OpenAI but missed by Perspective, compared to only 53 flagged by Perspective but missed by OpenAI. These preliminary numbers suggest Perspective may be substantially less sensitive to the kinds of content present in polarized communities, though these reflect raw API thresholds and not yet the human-annotated gold standard.

3.3 Next Steps

The following work remains: (1) complete OpenAI Moderation scoring on normalized text (1,316 items remaining)

(2) conduct annotator calibration session and re-annotation of revised 50-item sample (3) compute Cohen’s kappa, adjudicate disagreements, expand to 100+ items if $\kappa \geq 0.6$ (4) run detection gap analysis (precision/recall/F1 for all four conditions: Perspective raw, OpenAI raw, Perspective+normalization, OpenAI + normalization) (5) break down results by evasion strategy type; (6) write final report with visualizations. Steps (1) and (2) will proceed in parallel through the week of April 14, with step (3) completing by April 21; steps (4)-(6) are scheduled for the final two weeks before the May 4 presentation.

4 Anticipated Challenges or Risks

Dogwhistle Boundary: The whole point of a dogwhistle is that it can pass as normal speech. Most terms on any dogwhistle glossary are words that are used innocently far more often than they are used hatefully. Our codebook must establish clear decision rules when it comes to posts where dogwhistle signals are unclear. When this happens during the establishment of our human-annotated gold standard, posts will be assigned to a "Borderline/Ambiguous" category. In our detection gap analysis, Borderline/Ambiguous items will be excluded from the primary precision and recall calculations and reported separately, so that our core findings reflect only cases where human annotators reached clear agreement. We will also report the proportion of items classified as Borderline/Ambiguous as a measure of the inherent difficulty of covert hate speech classification, since it is a finding that is itself relevant to the Trust and Safety discussion as it quantifies the gray zone that both human moderators and automated systems have to navigate.

Lexicon Completeness: Conventionalization of existing hate speech, abandonment of algorithmically detected terms, and repurposing of neutral words into dogwhistles are nonstop processes in online hate communities. What we define as a dogwhistle lexicon is a snapshot in time and not a comprehensive glossary. We will document the construction process of our lexicon in full, so that we can properly report our data collection time frame in our final write-up. We expect this to be one of the trickier parts of the project. Our lexicon will inevitably miss terms that emerge after we compile it, and may include terms that have fallen out of use. This is a known limitation, not something we can fully solve.

Selection bias: The subreddits we choose to sample from will determine what evasion strategies and language patterns we observe. We sampled from a variety of subreddits, from hate-adjacent communities to intensely moderated and relatively neutral spaces like r/science. We documented our subreddit selection criteria transparently. Our findings can be interpreted as a characterization of detection gaps in the communities we sampled from, and not as a universal

account of covert hate speech on Reddit as a whole. If we only look at the worst corners of Reddit, we will find a lot of covert hate speech and overestimate the problem. Conversely, if we only look at well-moderated spaces, we will find almost none. The mix matters, and we will be transparent about it.

Inter-Rater Reliability: A risk we have already encountered is annotator calibration failure. Our initial pilot annotation round produced Cohen’s $\kappa = 0.00$, because one annotator classified all items as “Not Hateful” while the other identified the majority as hate speech. This was caused partly by a sample design issue: the initial pilot was stratified around API agreement and disagreement rather than direct enrichment for covert-hate prevalence, which resulted in many benign items and made annotator calibration more difficult. We have addressed this by redesigning the sample using multi-signal scoring with stratification by target group to ensure broader representation of potentially covert hate-bearing content.

Annotator Wellbeing: Sustained exposure to hateful content poses psychological risks to team members. We manage this through structured annotation shifts with mandatory breaks, a two-hour cap per annotation session, and regular team check-ins.

API Rate Limits: OpenAI Moderation quota limits daily batch throughput, slowing our scoring of the normalized dataset. We are using async batch submission to maximize throughput within quota; approximately 1,316 items remain.

Annotation Quality: If the revised pilot still produces $\kappa < 0.6$, we will conduct a second calibration round with more explicit codebook examples and consider collapsing the “Borderline/Ambiguous” and “Not Hateful” categories to reduce annotator uncertainty.

5 Expected Outcomes

By the end of the semester, our team expects to deliver the following:

- A human-annotated dataset of 100-200 Reddit posts and comments labeled for overt hate, covert hate, borderline, and non-hateful content, with evasion strategy annotations.
- A taxonomy of evasion strategies observed in Reddit communities, with frequency counts, representative examples (anonymized and paraphrased), and an assessment of each strategy’s effectiveness at evading Perspective API and OpenAI’s Moderation API.
- A quantified detection gap analysis reporting precision, recall, and F1 for each off-the-shelf classifier on overt

versus covert hate speech, showing the magnitude of the gap.

- A custom normalization and lexicon pre-processing pipeline (open-source Python code) with documented improvement in detection rates when paired with the same off-the-shelf classifiers.
- A final report synthesizing findings, situating them within the Trust and Safety literature on adversarial content moderation, and offering evidence-based recommendations for how platform moderation teams could address the covert hate speech gap, including specific technical interventions and policy considerations.
- A presentation summarizing our methodology, key findings, and recommendations

Specifically, we intend to recommend: (a) that platforms integrate a two-stage pre-processing pipeline before moderation API calls – first Unicode normalization and leetspeak decoding, then dogwhistle flag escalation for human review rather than automated action; (b) a human-in-the-loop protocol for items where dogwhistle flags co-occur with borderline toxicity scores, given the high false-positive rate of lexicon matching, and (c) that API providers such as Google consider adding coded-language subscales (e.g., an “identity_attack_coded” dimension) that score items for coded rather than explicit derogatory language.

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